

Waves and Thermodynamics Solutions

Question 1 (Conceptual)

The speed of sound in an ideal gas, like air, is directly related to the absolute temperature. The relationship is approximately given by:

$$v \propto \sqrt{T}$$

where T is the absolute temperature (in Kelvin).

Since 30°C is a higher temperature than 0°C , the sound waves propagating through the 30°C air will travel **faster**. This is because the molecules in the warmer air are moving faster and can therefore transmit the sound energy (the mechanical vibration) more quickly.

Question 2 (Calculation)

Given values:

- Speed of sound in helium at 20°C (v_1): $1,007\text{ m/s}$
- Distance (d): $2.5\text{ km} = 2,500\text{ m}$

(a) Write down the equation relating speed, distance, and time.

The basic relationship for constant speed is: $v = \frac{d}{t}$

(b) Calculate the travel time of the pulse.

Rearrange the equation to solve for time (t): $t = \frac{d}{v}$

Substitute the known values:

$$t_1 = \frac{2,500 \text{ m}}{1,007 \text{ m/s}}$$

The travel time is: $t_1 \approx 2.48 \text{ s}$

(c) If the temperature increases to 100 °C and the speed of sound in helium increases by 10%, what is the new travel time over the same distance?

First, calculate the new speed (v_2):

$$v_2 = v_1 + 0.10 \cdot v_1 = 1.10 \cdot v_1$$

$$v_2 = 1.10 \cdot 1,007 \text{ m/s}$$

The new speed is: $v_2 = 1,107.7 \text{ m/s}$

Now, calculate the new travel time (t_2) over $d = 2,500 \text{ m}$:

$$t_2 = \frac{d}{v_2}$$

$$t_2 = \frac{2,500 \text{ m}}{1,107.7 \text{ m/s}}$$

The new travel time is: $t_2 \approx 2.26 \text{ s}$